JavaCPUtils: FastIO API Cheat Sheet

High-speed Input & Output Reference for Competitive Programming.

Up to 10x faster reading than `Scanner`, 30x faster writing than `System.out.println`.

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1. Quick Start: The 10-Second Contest Template

```
import cp.*;

public class Solution {
    public static void main(String[] shd) {
        int t = FastIO.nextInt();
        while (t-- > 0) {
            int n = FastIO.nextInt();
            int[] arr = FastIO.nextIntArray(n);     // Read array in 1 line!
            IntPair edge = FastIO.nextIntPair();   // Read pair in 1 line!

            // Solve problem...

            FastIO.println("YES");
            FastIO.println(arr);                    // Prints 10 20 30 with newline!
            FastIO.println(edge);                   // Prints (u, v) with newline!
        }
        FastIO.flush(); // CRITICAL: Always flush at the end of main()!
    }
}
```

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2. Complete Fast Input Methods (`FastIO` / `FastScanner`)

Operation	Method Signature	Return Type	Description
Check Available	`FastIO.hasNext()`	`boolean`	Returns `true` if there are more non-whitespace tokens.
Read String Token	`FastIO.next()`	`String`	Reads next whitespace-delimited word.
Read Entire Line	`FastIO.nextLine()`	`String`	Reads remainder of line until `\n`.
Read 32-bit	`FastIO.nextInt()`	`int`	Reads next integer. Handles negative signs.

Integer			
Read 64-bit Long	`FastIO.nextLong()`	`long`	Reads next 64-bit integer up to 10^{18}.
Read Decimal	`FastIO.nextDouble()`	`double`	Reads next floating-point number.
Read Integer Array	`FastIO.nextIntArray(n)`	`int[]`	Reads `n` space/newline-separated ints into an array.
Read Long Array	`FastIO.nextLongArray(n)`	`long[]`	Reads `n` space/newline-separated longs into an array.
Read IntPair	`FastIO.nextIntPair()`	`IntPair`	Reads next two ints directly into an `IntPair`.
Read LongPair	`FastIO.nextLongPair()`	`LongPair`	Reads next two longs directly into a `LongPair`.

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3. Complete Fast Output Methods (`FastIO` / `FastWriter`)

All `println()` methods automatically append a newline (`\n`). No manual `\n` needed!

Operation	Method Signature	Trailing Newline?	Description
Print Value	`FastIO.print(val)`	❌ No	Writes `int`, `long`, `double`, `char`, `String`, or `Object`.
Print with Newline	`FastIO.println(val)`	✅ Yes	Writes value followed by `\n`.
Print Blank Line	`FastIO.println()`	✅ Yes	Writes an empty line (`\n`).
Print Array (1 Line)	`FastIO.println(int[] a)`	✅ Yes	Prints elements separated by space + `\n` at the end.
Print Array Inline	`FastIO.print(int[] a)`	❌ No	Prints elements separated by space without trailing `\n`.
Print Long Array	`FastIO.println(long[] a)`	✅ Yes	Prints long array space-separated + `\n` at the end.
Print Pair	`FastIO.println(pair)`	✅ Yes	Prints `(first, second)` followed by `\n`.

Flush Buffer	`FastIO.flush()`	—	**MANDATORY**: Dumps 64KB RAM buffer to terminal/judge.

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4. Key Rules to Remember in Live Contests ⚠️

- **Always Call `FastIO.flush()` at the end of `main()`:**

Because `FastWriter` stores output in high-speed RAM, if you forget `FastIO.flush()`, your output may stay trapped in memory and Codeforces will show **Wrong Answer** on empty output!

- **Online Judge Submissions:**

Run `bundle Solution.java` before uploading to Codeforces, LeetCode, CSES, or HackerRank.